

Rishi Raval

+91 87800 49045 | rishipraval@gmail.com | [imrishi007](https://github.com/imrishi007) | [rishipraval07](https://www.linkedin.com/in/rishipraval07) | rishipraval.vercel.app | Ahmedabad, India

EDUCATION

Adani University

B.Tech in Computer Science (AI-ML) | CGPA: 8.69 / 10.0

Ahmedabad, Gujarat

Aug 2023 – Present

- **Coursework:** Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Operating Systems, Theory of Computation, Computer Networks, DBMS

American School of Baroda

Higher Secondary Education

Baroda, Gujarat

Graduated May 2023

TECHNICAL SKILLS

Languages C++, C, Python, Java, JavaScript, SQL **Web:** React, HTML/CSS

Quant & ML PyTorch, CUDA, NumPy, Pandas, Scikit-learn, SciPy, Matplotlib, yfinance

Tools / OS Git, Linux (Fedora, Kali, Arch), VS Code, Jupyter Notebooks

PROJECTS

Multimodal Volatility & Direction Forecasting *PyTorch · FinBERT · GAT · CNN-BiLSTM · CUDA*

[GitHub](#) ↗

- Built a **gated fusion model** over 30 tech stocks combining price sequences (CNN-BiLSTM), SEC 10-K filings (FinBERT), a 30-node sector graph (GAT), and macro features (VIX, yields, credit spreads) for joint volatility and direction forecasting.
- Achieved **volatility** $R^2 = 0.921$ via QLIKE-primary loss, HAR-RV autoregressive features, and a skip connection bypassing the CNN-BiLSTM bottleneck—closing 95.7% of the gap to the HAR-RV econometric benchmark.
- Custom **GAT in pure PyTorch** over supply-chain and competitor edges yielded **+2.6% AUC gain**. Self-caught a critical **data leakage** (forward-looking peer returns) that collapsed AUC from ~ 0.94 to realistic levels after correction.
- Direction strategy (AUC 0.591) achieved **Sharpe 0.62 at 5 bp** transaction costs; weekly long-short backtest break-even ≈ 10 bp across 30 stocks.

FastGreeks — High-Performance Options Library *C++20 · BSM · Templates*

[GitHub](#) ↗

- Header-only **C++20 library** for Option Greeks (Delta, Gamma, Vega, Theta, Rho) and IV solving via Newton-Raphson with Bisection fallback; achieves **8.1 B pricing calls/sec** at -O3 with explicit edge-case handling for $T \rightarrow 0$, $\sigma \rightarrow 0$, and extreme moneyness.
- Polymorphic `CallOption`/`PutOption` interface grounded in first-principles derivation of the BSM PDE and Itô's Lemma; referenced Black & Scholes (1973) and Merton (1973) directly.

AlphaVision — Stock Direction Predictor & Dashboard *XGBoost · Streamlit · Scikit-learn*

[GitHub](#) ↗

- Cross-ticker **XGBoost classifier** on 76 normalised technical indicators achieving **60.2% walk-forward accuracy** across 8 equities; evaluated 28 configurations over 4 algorithms and 3 prediction horizons.
- Deployed as a live **Streamlit dashboard** with real-time Yahoo Finance data, interactive candlestick charts, and probability-calibrated direction predictions.

Quantitative Backtesting Engine *Python · Pandas · Vectorized Backtesting*

[GitHub](#) ↗

- Vectorized framework for SMA/RSI mean-reversion and momentum strategies with Sharpe, Sortino, and Max Drawdown metrics; grid-search parameter optimization with realistic cash-limited position sizing.

CERTIFICATIONS & ACHIEVEMENTS

JPMorgan Chase & Co. — Quantitative Research Virtual Experience | *Forage*

Nov 2025

- Modeled credit default probability from loan book data; used dynamic programming to bucket FICO scores and visualize credit risk distributions.

Competitive Programming | *LeetCode*

- **550+ problems** solved across Easy, Medium, and Hard. Contest Rating: **1444**.

INTERESTS

Research Deep learning for asset pricing, graph-based financial networks, volatility modeling, alternative data in systematic strategies, ML-in-finance (Gu/Kelly/Xiu, Cont)

Algorithms Competitive programming, dynamic programming, graph theory, combinatorics — drawn to reasoning-heavy problems as much as to quant

Building Exploring a unified finance platform combining quant tools, market data, and news for practitioners and retail investors

Writing [Recursify](#) & [Rishi Writes](#) | Linux, Photography